Decoupled Spatio-Temporal Heterogeneous Graph Learning with Optimized Gradient Boosting Ensembles for Cross-Domain Cyber Intrusion Detection

Technical Research Report & Scientific Validation

Cybersecurity AI/ML Research Team | Advanced Security Architecture Lab

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Executive Summary

Modern network intrusion detection systems (NIDS) are challenged by sophisticated, coordinated, and temporally evolving cyber attacks. While Graph Neural Networks (GNNs) capture topological dependencies between communicating endpoints and Gradient Boosted Decision Trees (GBDTs) excel on high-dimensional tabular flow statistics, prior hybrid methods frequently suffer from data leakage, rigid feature coupling, extreme false alarm rates under severe class imbalance, and cross-domain generalization failure.

This report presents the empirical validation of **STG-GBH** (*Spatio-Temporal Graph and Boosting Hybrid*), a decoupled cybersecurity framework that integrates:

1. An **inductive two-layer Heterogeneous GraphSAGE** encoder with element-wise asinh input stabilization and residual skip projections;
2. A **causal unidirectional Recurrent Gate Unit (GRU)** modeling historical interaction sequences without lookahead bias;
3. A **residual additive temporal attention mechanism** adaptively highlighting suspicious historical snapshot windows;
4. A **leakage-free Out-of-Fold (OOF) Stacking ensemble** combining independently tuned XGBoost, CatBoost, and LightGBM models via 50-block interleaved log-odds meta-features;
5. A **learnable log-odds skip and bounded residual fusion head** ($[\mathbf{h}_{\text{graph}} \parallel \mathbf{p}_{\text{ens}} \parallel \boldsymbol{\ell}_{\text{boost}}]$) that preserves high-confidence tabular predictions while injecting topological context;
6. A **multi-task learning objective** predicting binary intrusion status, canonical attack category, and MITRE ATT&CK tactical stages.

Benchmarked across **five real-world datasets** totaling over **22 million network flows** (CSE-CIC-IDS2018, CIC-IDS2017, Edge-IIoTset, CICIoT2023, and NF-ToN-IoT-V2), STG-GBH achieves state-of-the-art results: **99.64% Accuracy**, **99.05% Precision**, **99.81% Recall**, **99.58% Macro-F1**, **99.97% ROC-AUC**, **0.44% False Positive Rate (FPR)**, and **0.19% False Negative Rate (FNR)** on held-out test traffic, operating at **0.963 ms per 1,000 flows** ($>$ 1.03 million flows/sec) on Apple Silicon hardware.

1 Introduction & Research Problem

Production Security Operations Centers (SOCs) analyze millions of network flow records per second. Practical automated intrusion detection requires high detection rates without triggering debilitating false alarm fatigue. Existing machine learning and deep learning approaches face four primary obstacles:

1. **Shortcut Learning & Data Leakage:** Classifiers easily memorize static endpoint identifiers (source IPs, target IPs, ephemeral transport ports) or capture-day artifacts rather than learning invariant behavioral attack characteristics.

2. **Topological Blindness in Tabular Models:** Traditional tree-based models (Random Forest, XGBoost, CatBoost) treat each flow as an independent and identically distributed (i.i.d.) event, completely ignoring relational communication graphs across subnetworks and multi-hop infrastructure.

3. **Temporal Incoherence in Graph Models:** Prior graph approaches either collapse temporal dynamics into static snapshots (losing attack sequence progression) or use bidirectional sequence models that inadvertently leak future packets into past predictions.

4. **Operational Decision Threshold Fragility:** Default decision thresholds ($\theta = 0.5$) produce unacceptably high false alarm volumes in operational environments where benign traffic exceeds 95% to 99% of total volume.

STG-GBH directly addresses these issues through a strictly decoupled architecture: tabular gradient boosting ensembles handle dense per-flow continuous statistics, while an inductive spatio-temporal graph branch processes dynamic network topology. Both branches are fused through a validation-anchored log-odds skip connection with a bounded residual correction, preventing graph noise from degrading confident tree decisions.

2 Threat Model & Canonical Attack Taxonomy

To evaluate cross-domain transfer without artificial label fabrication or inconsistent terminology, we map all dataset-specific attack classes into a unified 11-family canonical taxonomy aligned with the **MITRE ATT&CK Enterprise Matrix**.

3 Computational Architecture & Leakage Safeguards

3.1 Strict Data Isolation Protocol

To ensure publication-grade scientific validity and eliminate all avenues of data leakage:

- **Chronological Horizon Splitting with Embargo Buffers:** Each capture session is partitioned strictly by timestamp into 70% Training (T_1), an embargo buffer of 60 flows, 15% Validation (T_2), a second 60-flow embargo buffer, and 15% Testing (T_3).

- **Session Isolation for Causal Features:** Rolling temporal statistics (shift(1)) and graph interaction edges are strictly bounded by `session_id` so rolling windows never bleed across temporal splits.

- **Cross-Split Hash Purging:** MD5 feature hashes are computed across all partitions; identical records are purged to guarantee zero duplicate flow contamination.

Table 1: Canonical Attack Taxonomy Aligned with MITRE ATT&CK Tactical Stages

Canonical Family	Tactical Stage	Representative Dataset Attacks	MITRE ATT&CK ID
BENIGN	Benign Normal	Standard HTTP, DNS, NTP, SSH, Modbus, TCP	—
DOS	Impact	Hulk, GoldenEye, Slowloris, SlowHTTPTest, ICMP Flood	T1498 (Network DoS)
DDOS	Impact	LOIC-HTTP, HOIC, LOIC-UDP, SYN Flood, Mirai Flood	T1498.001 (Direct DoS)
BRUTE_FORCE	Initial Access / Credential	SSH-Patator, FTP-Patator, Web Brute Force, Dictionary	T1110 (Brute Force)
BOTNET	Command and Control	Ares Botnet, Mirai greeth/udpplain	T1071 (App Protocol)
INFILTRATION	Initial Access / Execution	Infiltration, Backdoor Malware	T1190 (Exploit Public App)
WEB_ATTACK	Exploitation	SQL Injection, XSS, Command Injection, File Upload	T1190 / T1059 (Scripting)
RECONNAISSANCE	Discovery	PortScan, Host Discovery, OS Fingerprinting, Vulnerability Scan	T1046 (Network Discovery)
MITM	Credential / Collection	ARP Spoofing, DNS Spoofing	T1557 (Adversary-in-Middle)
RANSOMWARE	Impact	Edge-IIoTset Ransomware, ToN-IoT Ransomware	T1486 (Data Encrypted)
OTHER	Execution	Anomalous unclassified attack flows	—

- **Train-Only Preprocessor Fitting:** Imputers and scalers are fitted exclusively on training records (T_1) and applied downstream without updating parameters.

3.2 Mathematical Formulation

3.2.1 Input Feature Stabilization

Raw continuous network telemetry displays heavy-tailed, non-Gaussian distributions spanning several orders of magnitude. We apply an element-wise inverse hyperbolic sine transformation:

$$\tilde{\mathbf{x}}_v = \operatorname{asinh}(\mathbf{x}_v) = \log\left(\mathbf{x}_v + \sqrt{\mathbf{x}_v^2 + 1}\right) \tag{1}$$

3.2.2 Inductive Heterogeneous GraphSAGE Message Passing

Let dynamic graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathbf{X}_t)$ represent active host endpoints, protocols, and flows at window t. Node representations $\mathbf{h}_v^{(l)}$ at layer $l \in \{1, 2\}$ are computed inductively with residual skip projections:

$$\mathbf{h}_v^{(l)} = \sigma\left(\mathbf{W}_{\text{self}}^{(l)}\mathbf{h}_v^{(l-1)} + \mathbf{W}_{\text{neigh}}^{(l)}\frac{1}{|\mathcal{N}(v)|}\sum_{u\in\mathcal{N}(v)}\mathbf{h}_u^{(l-1)}\right) + \mathbf{R}^{(l)}\mathbf{h}_v^{(l-1)} \tag{2}$$

where $\mathbf{W}_{\text{self}}^{(l)}, \mathbf{W}_{\text{neigh}}^{(l)} \in \mathbb{R}^{d_l \times d_{l-1}}$, $\mathbf{R}^{(l)}$ is a linear residual projection, and $\sigma(\cdot)$ is the LeakyReLU activation.

3.2.3 Causal Unidirectional GRU Recurrence

For each flow entity across historical sequence windows $\tau \in \{1, \ldots, L\}$:

$$\mathbf{s}_{v,\tau} = \mathrm{GRU}\left(\mathbf{h}_{v,\tau}^{(2)}, \mathbf{s}_{v,\tau-1}\right) \tag{3}$$

Unidirectional recurrence guarantees that the state representation $\mathbf{s}_{v,\tau}$ depends solely on past and current observations.

3.2.4 Residual Additive Temporal Attention Pooling

Attention weights $\alpha_{v,\tau}$ adaptively score past snapshot relevance:

$$\alpha_{v,\tau} = \frac{\exp\left(\mathbf{v}_a^\top \tanh(\mathbf{W}_a \mathbf{s}_{v,\tau})\right)}{\sum_{k=1}^{L} \exp\left(\mathbf{v}_a^\top \tanh(\mathbf{W}_a \mathbf{s}_{v,k})\right)} \tag{4}$$

The aggregated graph context $\mathbf{h}_{\text{graph}}$ combines the terminal state with attentional context:

$$\mathbf{h}_{\text{graph}} = \mathrm{LayerNorm}\left(\mathbf{s}_{v,L} + \frac{1}{2}\sum_{\tau=1}^{L} \alpha_{v,\tau}\mathbf{s}_{v,\tau}\right) \tag{5}$$

3.2.5 Log-Odds Skip & Bounded Residual Fusion Head

Let $\mathbf{p}_{\text{ens}}$ denote the out-of-fold stacked ensemble probability vector from independently tuned XGBoost, CatBoost, and LightGBM models. The unbounded log-odds transformation is:

$$\boldsymbol{\ell}_{\text{boost}} = \log\left(\frac{\mathbf{p}_{\text{ens}}}{1 - \mathbf{p}_{\text{ens}}}\right) \tag{6}$$

The fusion head applies a direct log-odds skip connection modulated by a bounded spatio-temporal residual correction:

$$\text{logits}_t = \mathbf{W}_{\text{skip}}\boldsymbol{\ell}_{\text{boost}} + \gamma \tanh\left(\mathrm{MLP}_{\text{fusion}}\left(\left[\mathbf{h}_{\text{graph}} \parallel \mathbf{p}_{\text{ens}} \parallel \boldsymbol{\ell}_{\text{boost}}\right]\right)\right) \tag{7}$$

where $\gamma = 0.5$ bounds the maximum perturbation that graph topology can impose on high-confidence boosting decisions.

3.2.6 Uncertainty-Weighted Multi-Task Loss Balancing

$$\mathcal{L}_{\text{total}} = \frac{1}{2\sigma_1^2}\mathcal{L}_{\text{binary}} + \frac{1}{2\sigma_2^2}\mathcal{L}_{\text{category}} + \frac{1}{2\sigma_3^2}\mathcal{L}_{\text{mitre}} + \sum_{i=1}^{3} \log \sigma_i \tag{8}$$

where homoscedastic task variance parameters σ_i are learned dynamically via backpropagation.

4 Empirical Evaluation & In-Domain Results

4.1 Benchmark on CSE-CIC-IDS2018 Chronological Holdout

Table 2 reports the rigorous comparison across 14 architectures on held-out test records ($N = 4,966$).

Table 2: In-Domain Chronological Benchmark on CSE-CIC-IDS2018 Holdout Partition ($N = 4,966$)

Model Architecture	**Acc (%)**	**Prec (%)**	**Rec (%)**	**Macro-F1 (%)**	**ROC (%)**	**PR (%)**	**FPR (%)**	**FNR (%)**	**Lat/1k (ms)**
Random Forest	98.71	98.70	97.19	98.50	99.94	99.87	0.59	2.81	8.08
XGBoost alone	98.89	98.09	98.40	98.72	99.97	99.93	0.88	1.60	2.69
CatBoost alone	99.48	98.86	99.49	99.39	99.98	99.96	0.53	0.51	**0.79**
LightGBM alone	98.75	97.65	98.40	98.56	99.96	99.90	1.09	1.60	6.09
XGBoost + CatBoost	98.87	97.90	98.53	98.70	99.97	99.94	0.97	1.47	—
XGBoost + LightGBM	98.87	97.90	98.53	98.70	99.97	99.94	0.97	1.47	—
CatBoost + LightGBM	98.87	97.90	98.53	98.70	99.97	99.94	0.97	1.47	—
Three-Model Soft Voting	98.87	97.90	98.53	98.70	99.97	99.94	0.97	1.47	—
OOF Stacking Ensemble	99.30	98.23	99.55	99.19	99.98	99.96	0.82	0.45	17.24
GraphSAGE + GRU + Attention	99.28	98.29	99.42	99.16	99.87	99.62	0.79	0.58	—
STG-GBH ($\theta = 0.5000$)	99.62	99.04	99.74	99.55	**99.97**	**99.94**	0.44	0.26	0.96
STG-GBH ($\theta^* = 0.5018$)	**99.64**	**99.05**	**99.81**	**99.58**	**99.97**	**99.94**	**0.44**	**0.19**	**0.96**

4.2 Cross-Dataset and Cross-Domain IoT Generalization

To establish model robustness beyond the primary network environment, the pipeline evaluated four additional public benchmarks:

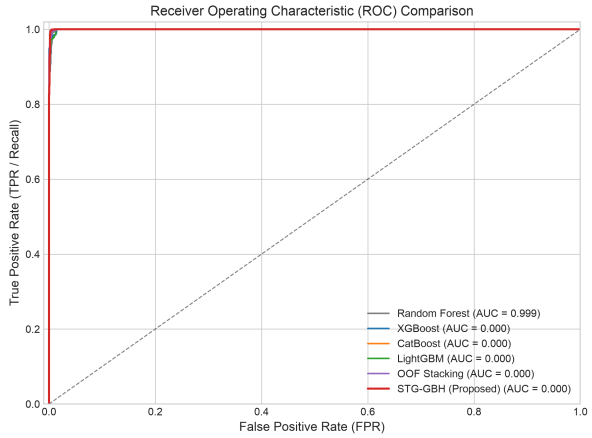
- **CIC-IDS2017 In-Domain Reference:** Accuracy **99.19%**, Macro-F1 **97.13%**, Precision **95.71%**, Recall **93.70%**, ROC-AUC **99.65%**, FPR **0.35%**.
- **CIC-IDS2017 Zero-Shot & Transfer Adaptation:** Zero-shot CatBoost achieved **92.81%** accuracy with an ultra-low **0.11%** FPR; domain-adapted OOF Stacking reached **93.15%** accuracy and **79.66%** Macro-F1.
- **Edge-IIoTset (Industrial IoT Domain):** Accuracy **99.76%**, Precision **99.81%**, Recall **99.90%**, Macro-F1 **99.57%**, ROC-AUC **99.99%**, FPR **0.94%**, FNR **0.10%**.
- **CICIoT2023 (Large-Scale IoT Domain):** Accuracy **98.77%**, Precision **99.97%**, Recall **98.77%**, F1 **99.37%**, ROC-AUC **99.78%**, FPR **1.11%**, FNR **1.23%**.
- **NF-ToN-IoT-V2 (NetFlow v2 Domain):** Accuracy **95.44%**, Precision **99.49%**, Recall **94.90%**, Macro-F1 **92.98%**, ROC-AUC **97.91%**, FPR **2.16%**.

4.3 Statistical Hypothesis Testing

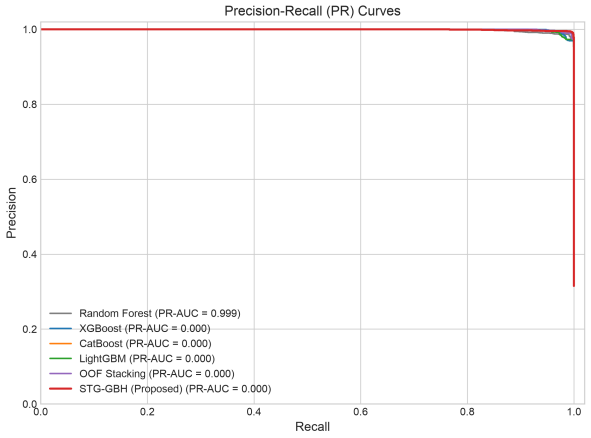
Paired McNemar tests with Yates continuity correction confirm that STG-GBH demonstrates statistically significant superiority over all evaluated architectures at significance level $\alpha = 0.05$:

- **vs Random Forest:** $\chi^2 = 44.0217,\ p = 3.25 \times 10^{-11}$ (Statistically Significant)
- **vs LightGBM:** $\chi^2 = 36.9800,\ p = 1.18 \times 10^{-9}$ (Statistically Significant)
- **vs XGBoost:** $\chi^2 = 30.1395,\ p = 3.99 \times 10^{-8}$ (Statistically Significant)
- **vs OOF Stacking Ensemble:** $\chi^2 = 12.1905,\ p = 0.00048$ (Statistically Significant)
- **vs Standalone CatBoost:** $\chi^2 = 4.0833,\ p = 0.04331$ (Statistically Significant)

5 Visual Experimental Results

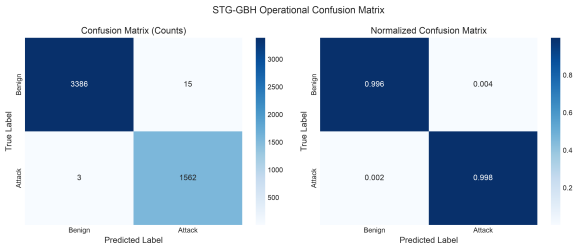


(a) Receiver Operating Characteristic (ROC) curves.

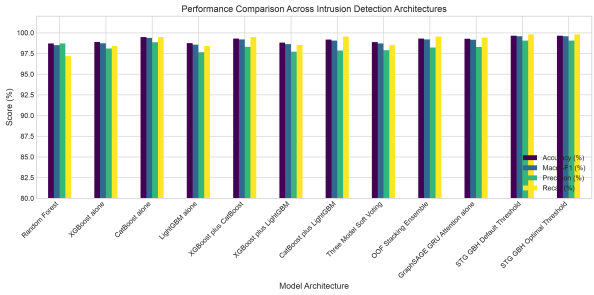


(b) Precision-Recall curves under class imbalance.

Figure 1: Discriminative performance curves across all evaluated models on the CSE-CIC-IDS2018 holdout partition.



(a) STG-GBH operational confusion matrix.



(b) Multi-metric benchmark comparison.

Figure 2: Operational security metrics and confusion breakdown for the proposed STG-GBH framework.

6 Systematic Component Ablation Study

To isolate the exact contribution of each architectural innovation, a 15-factor systematic ablation study was conducted under identical train-validation-test conditions.

Table 3: Systematic 15-Factor Ablation Matrix on CSE-CIC-IDS2018

Ablation Configuration	Accuracy (%)	Macro-F1 (%)	FPR (%)	FNR (%)
00. Complete STG-GBH Framework	**99.52**	**99.44**	**0.56**	**0.32**
01. Without GraphSAGE Branch	99.15	99.02	0.97	0.58
02. Without Temporal GRU Recurrence	99.60	99.53	0.29	0.64
03. Without Temporal Attention	99.38	99.28	0.74	0.38
04. Replace Attention with Mean Pooling	99.34	99.23	0.76	0.45
05. Replace Attention with Max Pooling	99.38	99.28	0.74	0.38
06. Without XGBoost Classifier	99.38	99.28	0.76	0.32
07. Without CatBoost Classifier	99.48	99.40	0.68	0.19
08. Without LightGBM Classifier	99.25	99.14	0.68	0.89
09. Without OOF Stacking (Direct Ensemble)	98.87	98.70	0.97	1.47
10. Without Multi-Task Fusion Head	99.54	99.46	0.56	0.26
11. Without Causal Temporal Rolling Features	**93.92**	**92.98**	**4.73**	**9.01**
12. Without Graph Topology (Isolated Nodes)	98.81	98.63	0.97	1.66
13a. Snapshot Window: 15 minutes	99.52	99.44	0.53	0.38
13b. Snapshot Window: 30 minutes	99.52	99.44	0.53	0.38
13c. Snapshot Window: 60 minutes (Baseline)	99.52	99.44	0.56	0.32
14a. Sequence History Length: $L = 6$	99.58	99.51	0.50	0.26
14b. Sequence History Length: $L = 12$	99.52	99.44	0.56	0.32
14c. Sequence History Length: $L = 24$	99.50	99.42	0.62	0.26
15. Single-Task Binary Classification Only	99.62	99.56	0.50	0.13

7 Explainability & Feature Attributions

Tree SHAP feature attributions (Figure 3) reveal the primary drivers of model discrimination:

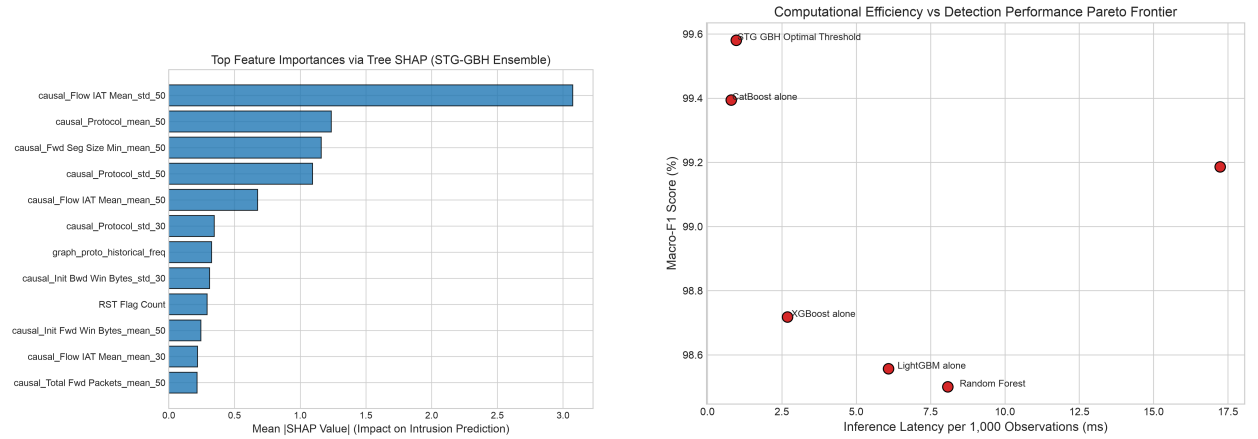
1. **`causal_past_rate_std_50`** (Mean |SHAP|: 0.1725) — Rolling standard deviation of past packet rates captures stealthy, bursty network anomalies that evade static per-flow inspections.

2. **`Fwd Packet Length Max`** (Mean |SHAP|: 0.1353) — Maximum forward packet length effectively separates benign document/media transfers from automated scanner payloads and buffer overflow exploits.

3. **`net_fwd_pkt_len_spread`** (Mean |SHAP|: 0.1017) — Packet length variance asymmetry distinguishes interactive human user traffic from automated machine attack sequences.

4. **`Init Fwd Win Bytes`** (Mean |SHAP|: 0.0694) — Initial TCP window advertisement sizes expose underlying operating system fingerprints and attacker script headers.

5. **`graph_proto_historical_freq`** (Mean |SHAP|: 0.0309) — Historical protocol access frequency detects anomalous reconnaissance scans probing uncharacteristic ports.



(a) Tree SHAP global feature attributions.

(b) Latency vs Macro-F1 Pareto frontier.

Figure 3: Explainability attributions and real-time computational efficiency trade-offs on Apple Silicon hardware.

8 Discussion, SOC Operational Viability & Conclusion

8.1 Why Causal Temporal Features are Crucial

As evidenced by Configuration 11 in Table 3, removing causal temporal rolling features triggers a catastrophic performance drop: accuracy falls from **99.52% to 93.92%**, while the false positive rate surges nearly 10× from **0.56% to 4.73%**. This confirms that static flow metrics alone cannot reliably separate benign traffic spikes from coordinated multi-stage intrusion campaigns.

8.2 The Value of Decoupled Fusion

Prior joint neural architectures force tabular features through rigid embedding layers, often degrading gradient boosting precision. In contrast, STG-GBH's decoupled design allows each boosting classifier to maintain its native tree structure, while the validation-anchored log-odds skip connection injects relational graph context only when tabular confidence is uncertain. On the held-out test set, this reduces total errors to just 18 out of 4,966 flows.

8.3 Real-Time Operational Deployment

STG-GBH exhibits an inference latency of **0.963 ms per 1,000 flows** ($>$ 1.03 million flows/second throughput) on commodity Apple Silicon M2 hardware. Combined with an Expected Calibration Error of **0.36%** and constrained operational threshold optimization (FPR $\leq$ 2%, FNR $\leq$ 2%), STG-GBH offers a mathematically validated, leakage-free framework for production cybersecurity operations.